MILITARY SPECIFICATION

TRANSMISSION, CROSS-DRIVE (140 TO 205 HORSEPOWER)

1. SCOPE

1.1 Scope. - This specification covers two types of transmissions designed for use with engines ranging from 140 to 205 gross horsepower.

1.2 Classification. - Transmissions shall be of the following types:

Type I - XT-90-2
Type II - CD-150-4

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS
Military
JAN-P-100 - Packaging and Packing for Overseas Shipment, General Specification.
MIL-P-116 - Preservation, Methods of.
MIL-L-2104 - Lubricating Oil, Internal-Combustion Engine, Heavy-Duty.
MIL-L-10295 - Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.

ENGINEERING PARTS LIST
8348000 - Transmission XT-90.
8349157 - Transmission CD-150-4.

PUBLICATIONS
Packaging Data Sheets
8348000 - Transmission Assy. XT-90.
8349157 - Transmission Assy. CD-150-4.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Preproduction model. - The transmissions furnished under this specification shall be products which have been represented by a preproduction model that has been inspected and has passed the preproduction inspection specified herein (see 4.3 and 6.1).

3.2 Design and construction. - The design and construction shall be to the form and dimensions specified on detail drawings shown on the Engineering Parts List 8348000 or 8349157 as applicable.

3.2.1 Transmission oil. - Transmission oil conforming to grade 1 of Specification MIL-L-21260 shall be used during transmission testing and production runs except, that oil conforming to Specification MIL-L-10295 shall be used when transmission is operated in ambient temperatures below 0°F.

3.2.1.1 Service. - Transmission oil conforming to grade 10 of Specification MIL-L-2104 shall be used after installation of the transmission in its intended vehicle.

3.3 Performance.

3.3.1 Pressure requirements. - When tested as specified in 4.6.2, the transmissions shall meet the requirements specified in 3.3.1.1.

3.3.1.1 Torque and pressure allowance. - When tested as specified in 4.6.2.1, torque and oil pressures shall conform to the requirements specified in table I:

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Table I. Torque and pressure allowance
(At 1500 R.P.M.)

Item	Type I Min.	Type I Max.	Type II Min.	Type II Max.
Input torque - lbs.-ft.				35
Main oil pressure - psi	120	200	170	190
Converter "in" oil pressure - psi	65	95	85	
Lubricating oil pressure - psi	20		25	
Right hand and left hand output clutch oil pressure - psi	To be within 10 psi of main pressure			
First, second, third, and reverse gear clutch oil pressure - psi	To be within 10 psi of main pressure			
Right hand and left hand steer clutch oil pressure - psi	To be within 10 psi of main pressure			
Right hand and left hand shifting valve supply pressure - psi	To be within 10 psi of main pressure			
Right hand and left hand steer lever travel to drop steering clutch in high and low gear (degrees of travel) -			$6.7^\circ \pm 2^\circ$	
Right hand and left hand steer lever travel to apply brakes when in low gear, and low clutch in high range, (degrees of travel) -			$6.5^\circ \pm 3^\circ$	
Right hand and left hand brake lever travel to drop output clutch	functional			
Additional lever travel to apply right and left hand brake	functional			

3.3.2 Warm-up and service brake values. - When tested as specified in 4.6.3 through 4.6.3.2, the test results shall be within the limits specified in table II.

Table II. - Test values

Item	Type I Min.	Type I Max.	Type II Min.	Type II Max.
Input speed - rpm	2000	-	2000	-
Converter "out" oil temperature - °F.	200	225	200	225
Apply lever torque each side - lbs.-ft	45	-	50	-
Input torque - lbs.-ft. (Service brakes must not slip)	200	-	120	-

3.3.3 Lock-up clutch and governor values. - When tested as specified in 4.6.4 through 4.6.4.2, the test results shall be within the limits specified in table III.

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Table III. - Clutch and governor values

Item	Type I Min.	Type I Max.	Type II Min.	Type II Max.
Engine manifold vacuum at 0 inches of mercury				
Lock-up clutch engagement - rpm output speed	300	360	340	370
Lock-up clutch disengagement - rpm output speed	225	285	245	275
Engine manifold vacuum at 10 inches of mercury				
Lock-up clutch engagement rpm output speed	260	320		
Lock-up clutch disengagement - rpm output speed	175	235		

3.3.4 Load and no-load values. - When tested as specified in 4.6.5 through 4.6.5.6, the load and no-load values shall be within the applicable limits specified in table IV.

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Table IV. - Load and no-load values

*see acceptance limits Figure 1
**see acceptance limits Figure 2

Item		Load check Third gear stall	Third gear Mid-point	High gear stall	High gear Mid-point	No-load check Neutral range	First gear	Second gear	Third gear	Reverse gear	Neutral range	Low free run	High free run	Reverse free run	
		Type I	Type I	Type II	Type II	Type I	Type I	Type I	Type I	Type I	Type II	Type II	Type II	Type II	
1	Input speed r.p.m.	1950	2200	2600	2000	2600	2600	2600	2600	2600	2600	2600	2600	2600	
2	Output speed ± r.p.m.	0	220 / 270	0	210 / 260		96	180	406	134			430	186	
3	Input torque ± lbs. - ft.	*	*	**		15 Max.	20 Max.	23 Max.	25 Max.	20 Max.	30 Max.	35 Max.	35 Max.	30 Max.	
4	Output torque ± lbs. ± ft. (Min.)	*	*	**	**										
5	Main oil pressure - p.s.i.	190 215	100 Min.	170 190	170 190	65 Max.	75 Min.	75 Min.	75 Min.	75 Min.	170 190	170 190	170 190	170 190	
6	Converter "in" oil pressure - p.s.i.	60 85		40 110		65 Min.	65 Min.	65 Min.	65 Min.	65 Min.	95 125	95 125	100 130	100 120	
7	Converter "out" oil pressure - p.s.i.	50 Min.		50 100		55 Min.	55 Min.	55 Min.	55 Min.	55 Min.	40 65	40 65	40 65	40 65	
8	Converter "out" oil temperature - °F.	300 Max.		250 275		160 225	160 225	160 225	160 225	160 225	160 240	160 240	160 240	160 240	
9	Cooler oil flow - G.P.M.	8 Min.		12 Min.		11 Min.	12.5 Min.	14 Min.	14 Min.	12 Min.	9 Min.	9 Min.	9 Min.	6 Min.	
10	Lubricating oil pressure p.s.i.	20 30	20 Min.	20 40	20 Min.	25 Min.	25 Min.	25 Min.	25 Min.	25 Min.	25 Min.	25 Min.	25 Min.	20 Min.	
11	Right hand and left hand shifting valve supply pressure - p.s.i.			To be within 10 of main								To be within 10 of main			
12	First gear clutch apply pressure - p.s.i.	* **	* **	* **	* **	* **	80 psi of main	* **	* **	* **	* **	* **	* **	* **	
13	Second gear clutch apply pressure - p.s.i.							80 psi of main							
14	Third gear clutch apply pressure - p.s.i.	To be within 10 of main							80 psi of main						
15	Reverse gear clutch apply pressure - p.s.i.									80 psi of main					
16	Lock-up clutch oil pressure - psi.i.	80 psi of main									80 psi of main	80 psi of main	80 psi of main	80 psi of main	
17	Right hand and left hand Output clutch pressure - p.s.i.		80 psi of main												

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3.3.5 Type I transmission.

3.3.5.1 Shift inhibitor values. - When tested as specified in 4.6.6 and 4.6.6.1, the test results shall be within the limits specified in table V.

Table V. - Shift inhibitor values

Item	Value
Input speed - rpm (Inhibitor shall prevent downshift)	2100
Input speed - rpm (Inhibitor shall not prevent downshift)	1600
Input speed - rpm (Inhibitor shall prevent shift)	1860
Input speed - rpm (Inhibitor shall not prevent shift)	1050

3.3.6 Leak check. - When tested as specified in 4.6.7, the transmission shall show no evidence of leakage.

3.3.7 Temperature range. - When tested as specified in 4.6.8, the transmission shall operate satisfactorily throughout an ambient air temperature range of -65°F. to + 125°F. The oil temperature in the transmission shall not exceed 300°F.

3.4 Identification markings. - The manufacturers serial number shall be legible and permanently stamped into the metal of each major transmission component housing, in a location easily visible when the transmission is installed in its intended vehicle. The marking shall be in accordance with Military Standard MIL-STD-130.

3.5 Workmanship. - The workmanship shall be in accordance with high-grade manufacturing practice for the product covered by this specification and shall be such as to produce transmissions free of defects specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Contractor's quality assurance system. - The contractor shall provide and maintain an effective Inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced therein.

4.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor

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to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Preproduction model. - A preproduction model selected from the first two units of production (see 3.1), shall be submitted for inspection and testing as specified herein. The model shall be representative of the transmission proposed to be furnished under the contract. Preproduction inspection and testing shall be conducted by the Government at a place designated by the Government, and shall consist of visual examination and all tests specified in 4.6 through 4.6.8.

4.3.1 Failure. - Failure of a preproduction model to pass any test specified herein, shall be cause for refusal by the Government to conduct a retest until corrective measures, satisfactory to the Government, have been taken.

4.4 Acceptance inspection.

4.4.1 Sampling for transmission.

4.4.1.1 Lot formation. - Unless otherwise specified, a lot shall consist of all transmissions of one part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.1.2 Sampling for acceptance examination. - Samples for acceptance examination shall be selected in accordance with Standard MIL-STD-105.

4.4.2 Examination inspection for transmission.

4.4.2.1 Acceptable quality level. - Each model selected in accordance with 4.4.1.2, shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.4.2.2 Classification of defects. - For examination purposes, defects shall be classified as follows:

Major defects (1.0 AQL)	Examination method
Dimensions out of tolerance, affecting interchangeability of major components	SME (Standard Measuring Equipment)
Malfunction of control levers and valves	Functional & Visual
Improper installation of hydraulic lines and fittings	Visual
Improper linkage adjustment on controls	Visual
Dirt, foreign matter in oil filter	Visual
Oil leaks	Visual

Minor defects (2.5 AQL)	Examination method
Dimensions out of tolerance, not affecting interchangeability of major components	SME
Burs	Visual
Rust	Visual

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Scratches	Visual
Chips	Visual
Incorrect marking (see 3.4)	Visual

4.4.3 Acceptance tests. - Each transmission presented for acceptance shall be subjected to the tests specified in 4.6 through 4.6.3.

4.4.3.1 Failure. - Failure of a transmission to pass any specified test shall be cause for rejection of that transmission. Rejected transmission may be resubmitted for acceptance after objective evidence has been provided by the manufacturer that corrective action has been taken to eliminate the condition which caused rejection.

4.4.4 Sampling for packaging and packing.

4.4.4.1 Lot formation. - A lot shall consist of all packs prepared for shipment in accordance with one level (see 5.1), from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.4.2 Sampling for acceptance examination. - Samples for acceptance examination shall be selected in accordance with Standard MIL-STD-105.

4.4.4.3 Sampling for acceptance testing. - One complete pack and one package shall be selected for acceptance testing from each lot.

4.4.5 Examination inspection for packaging and packing.

4.4.5.1 Acceptable quality level. - Each sample selected in accordance with 4.4.4.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective.

Classification	AQL
Major	1.0
Minor	2.5

4.4.5.2 Classification of defects. - For examination purposes, defects shall be classified as follows:

Major defects (1.0 AQL)	Examination method
Improper preservative oil used in crankcase (see data sheet).	Visual
Openings improperly sealed (see data sheet).	Visual
Improper materials used in sealing the openings (see data sheet).	Visual
Unpainted exterior surfaces improperly processed (see data sheet).	Visual
Improper container being used (see data sheet).	Visual
Incorrect marking (see 5.1).	Visual
Illegible marking (see 5.1).	Visual

Minor defects (2.5 AQL)	Examination method
Improper closure and strapping of shipping container (see data sheet).	Visual

4.4.6 Testing inspection for packs.

4.4.6.1 Acceptance test. - When level A or B is specified (see 6.1), the sample selected in accordance with 4.4.4.3 shall be tested as specified in 4.7.1 or 4.7.2, as applicable.

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4.4.6.2 Failure. - Failure of the sample to pass any specified test, shall be cause for the Government to refuse to accept the lot until it has been proved to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.5 Pilot pack inspection.

4.5.1 Pilot pack. - The pilot pack (see 5.2) shall be examined for the defects in 4.4.5.2, and subjected to the tests specified in 4.7.1 and 4.7.2.

4.5.2 Rejection. - Failure of the pilot pack to pass any specified examination or test may be cause for refusal of the Government to conduct a retest, until it has been proven, to the satisfaction of the Government activity conducting the test, that the faults revealed by such tests have been corrected.

4.6 Test procedure.

4.6.1 Test conditions. - The transmission shall be installed on an absorption dynamometer test stand and all connections for input shafts, steering and shifting controls, oil lines, pressure and temperature outlets shall be made. The transmission output shafts shall be coupled to the dynamometer, with no load on the dynamometer. The transmission speed shall be increased to 1500 ± 15 r.p.m. until main and converter oil pressures are stabilized. Tests shall be conducted at room temperature.

4.6.2 Pressure tests. - To determine conformance to 3.3.1, the transmission shall be operated as specified in 4.6.2.1 and the readings recorded.

4.6.2.1 Torque and pressure allowance tests. - To determine conformance to 3.3.1.1, the transmission shall be operated through all steer and shift positions, and the following torques and pressures recorded:

(a) Input torque
(b) Main oil pressure
(c) Converter "in" oil pressure
(d) Lubricating "oil" pressure
(e) Right and left hand output clutch oil pressure (type I only)
(f) First, second, third, and reverse gear clutch oil pressure (type I only)
(g) Right and left hand steer clutch oil pressure (type II only)
(h) Right and left hand shifting valve supply pressure (type II only)
(i) Right and left hand steer lever travel to drop steering clutch in high and low gear (type II only)
(j) Right and left hand steer lever travel to apply brakes when in low and apply low clutch when in high range (type II only)
(k) Right and left hand brake lever travel to drop output clutch
(l) Additional lever travel to apply right and left hand brakes

4.6.3 Warm-up and service brake test. - With output shafts coupled to absorption dynamometer, the transmission shall be placed in third gear or high range and service brakes applied to hold third gear or high stall. Input speed shall be slowly increased to 2000 r.p.m. The service brakes shall not be permitted to slip. Warm-up shall be continued until all temperatures are stabilized. Conformance to 3.3.2 shall be determined.

4.6.3.1 Service brake type I. - With output shafts coupled to absorption dynamometer, the transmission shall be placed in second gear and the output shafts stalled by applying right and left hand service brakes with 45 lbs. - ft. torque on each apply lever. Input torque shall be slowly increased to 200 lbs. - ft. Conformance to 3.3.2 shall be determined.

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4.6.3.2 Service brake type II. - The test specified in 4.6.3.1 shall be repeated except the transmission shall be placed in high range with 50 lbs. - ft. torque on each apply lever and input torque slowly increased to 120 lbs. - ft. Conformance to 3.3.2 shall be determined.

4.6.4 Lock-up clutch engagement and disengagement. - For type I transmission with no load on output shafts and coupled to absorption dynamometer, a vacuum of 0 inches of mercury shall be simulated in engine manifold to transmission vacuum line. The transmission shall be placed in third gear and input speed slowly increased until lock-up clutch engages. Conformance to 3.3.3 shall be determined. The test shall then be repeated and lock-up clutch disengagement determined for conformance to 3.3.3.

4.6.4.1 Lock-up clutch engagement and test. - Type I transmission shall be operated as specified in 4.6.4, except that a vacuum of 10 inches of mercury shall be simulated in engine manifold to transmission vacuum line. Conformance to 3.3.3 shall be determined.

4.6.4.2 Lock-up clutch and governor test. - Type II transmission shall be operated as specified in 4.6.4, except that the transmission shall be placed in high gear range to determine conformance to 3.3.3.

4.6.5 Load and no-load tests.

4.6.5.1 Third gear stall. - The type I transmission output shafts shall be coupled to the absorption dynamometer and the transmission placed in third gear with service brakes released. The input speed shall be 1950 r.p.m., with output shafts stalled. Determine conformance to 3.3.4.

4.6.5.2 Third gear mid-points. - The type I transmission shall be operated as specified in 4.6.5.1, except that the input speed shall be 2200 r.p.m. and the output speed 245 ± 25 r.p.m. to determine conformance to 3.3.4.

4.6.5.3 High gear stall. - The type II transmission shall be operated as specified in 4.6.5.1, except that the transmission shall be placed in high gear and the input speed shall be 2600 r.p.m. with output shafts stalled to determine conformance to 3.3.4.

4.6.5.4 High gear mid-point. - The type II transmission shall be operated as specified in 4.6.5.1, except that the transmission shall be placed in high gear, the input speed shall be 2000 r.p.m. and the output speed 210 - 260 r.p.m. to determine conformance to 3.3.4.

4.6.5.5 Neutral range. - The type I transmission shall be operated as specified in 4.6.5.1, except that the transmission shall be placed in neutral range and the input speed increased to 2600 r.p.m. with no-load on the transmission output shaft. Conformance to 3.3.4 shall be determined through the following gear changes:

(a) First gear, (b) Second gear, (c) Third gear, and (d) Reverse gear.

4.6.5.6 Neutral test range. - The type II transmission shall be operated as specified in 4.6.5.1, except that the transmission shall be placed in neutral range with output shafts stalled and an input speed of 2600 r.p.m. to determine

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conformance to 3.3.4. This test procedure shall be repeated with no-load on transmission output shafts. Conformance to 3.3.4 shall be determined through the following gear changes: (a) Low free run, (b) High free run, and (c) Reverse free run.

4.6.6 Shift inhibitor test. - The type I transmission output shafts shall not be coupled to the absorption dynamometer. The transmission shall be placed in second gear with an input speed of 2100 r.p.m. Downshift into first gear shall be performed. Conformance to 3.3.5.1 shall be determined.

4.6.6.1 Downshift test. - The type I transmission shall be operated as specified in 4.6.6 through the following gear changes and input speeds. Conformance to 3.3.5.1 shall be determined.

(a) Second to first gear - 1600 r.p.m.
(b) First to reverse gear - 1860 r.p.m.
(c) First to reverse gear - 1050 r.p.m.

4.6.7 Leak check. - The transmission input speed shall be increased to 1000 r.p.m. with no-load on the transmissions output shafts. When the converter "out" oil temperature is between 270° - 300°F. and sump temperature is between 200° - 240°F., inspection shall be made for leaks, in all gear ranges, to determine conformance to 3.3.6.

4.6.8 Extreme temperature test. - To determine conformance to 3.3.7, the transmission shall be serviced with the applicable oil (see 3.2.5) and placed in an environmental chamber at a temperature of -65°F. ± 5°F., for a period of 24 hours. The transmission shall be coupled to the absorption dynamometer and all connections for output shafts, steering and shifting controls, oil lines, pressure and temperature outlets shall be made with the ambient temperature remaining at -65°F. ± 5°F., the transmission input speed shall be slowly increased to 1500 r.p.m. until main and converter oil pressures are stabilized. Check all steer and shift positions for any indication of noise, drag or malfunction. The transmission shall then be serviced with the applicable oil (see 3.2.1) and the above procedure shall be repeated except that the temperature shall be raised to 125°F. ± 5°F. The converter "out" oil temperature shall not exceed 300°F.

4.7 Packaging and packing tests.

4.7.1 Packaging test. - To determine conformance to 5.1.1, the unit package shall be subjected to the applicable tests of Specification MIL-P-116 for the method of unit protection specified on the applicable packaging data sheet (see 2.1).

4.7.2 Packing test. - To determine conformance to 5.1.2, the pack, packed in accordance with the requirements specified on the applicable packaging data sheet, shall be subjected to the interior packing tests of Specification JAN-P-100.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation, packaging, packing and marking. - Cleaning, preservation, packaging, packing and marking, for the level of protection specified (see 6.1), shall be in accordance with packaging data sheets 8348000 or 8349157 as applicable to the specific item by Government part number (see 2.1).

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5.1.1 Packaging. - When tested as specified in 4.7.1, the unit package shall meet the requirements for the method of unit protection of Specification MIL-P-116 specified on the applicable packaging data sheet.

5.1.2 Packing. - When tested as specified in 4.7.2, the packing specified on the applicable packaging data sheet shall meet the requirements for the interior packing tests of Specification JAN-P-100.

5.1.3 Material. - Material used in the packaging operation and specified on the appropriate packaging data sheet, shall conform to all requirements of the respective material specification.

5.1.4 Dimensions. - Basic dimensions shall be as specified on the applicable packaging data sheet. Tolerances are not specified; therefore, reasonable accuracy shall be maintained at all times with these dimensions specified and wherever possible commercial tolerances shall be utilized.

5.1.5 Container. - Data specified for exterior size, gross weight and cube will not be used for contractural purposes, since these figures are for logistics data only. Exterior container size will be the least that is required to contain the unit.

5.2 Pilot pack. - A pilot pack shall consist of one complete pack packaged as for shipment in accordance with the requirements of level A or B, specified on the applicable packaging data sheet (see 5.1). Prior to packing in quantity, the pilot pack shall be submitted by the contractor for inspection and testing as specified in 4.5.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When and where preproduction model is to be forwarded (see 3.1).
(c) Selection of the applicable level of preservation, packaging, and packing (see 5.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:
Army - Ordnance Corps

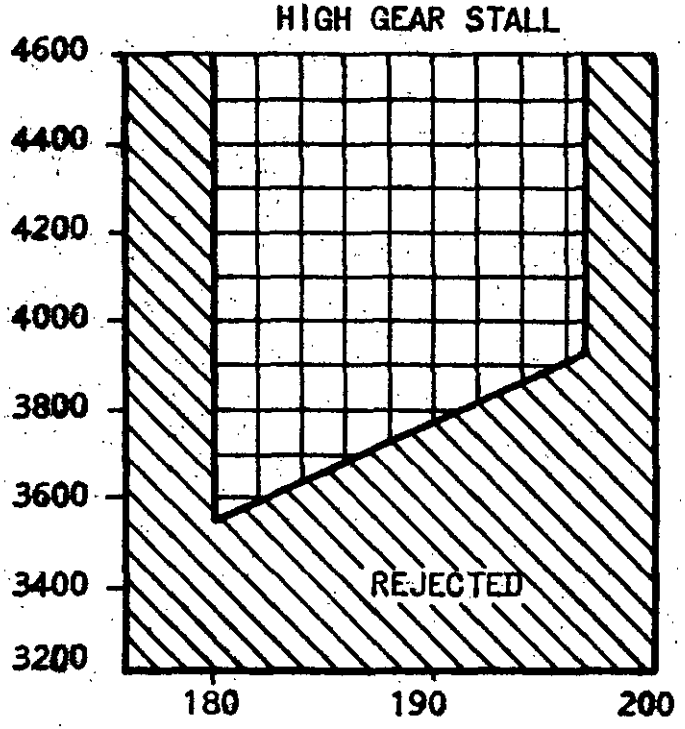
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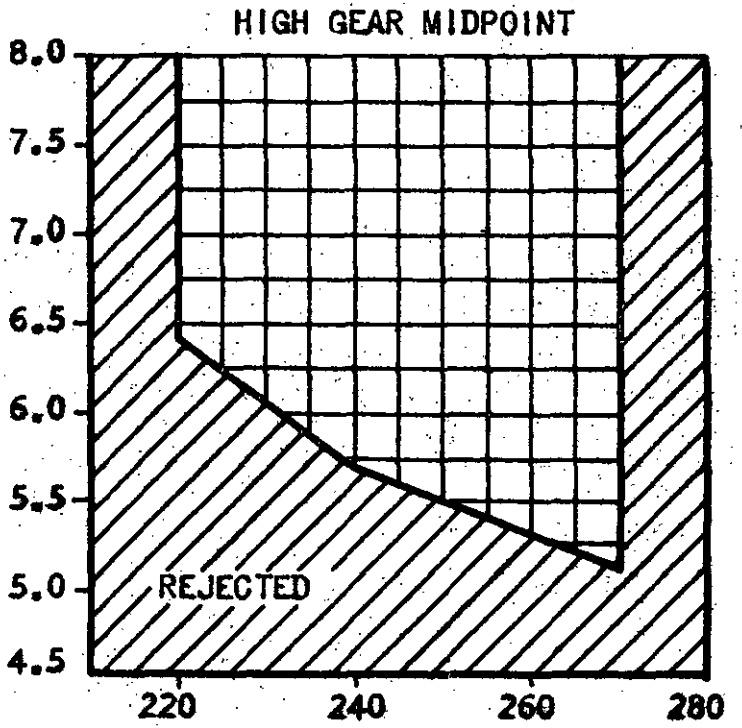
ACCEPTANCE LIMITS

FIGURE 1

GEAR RANGE	FREE RUN MAX. INPUT TORQUE	ACTUAL INPUT TORQUE
HIGH	25	
INT.	23	
LOW	20	
REV.	20	
NEU.	15	



TRANSMISSION INPUT TORQUE(FT-LB) AT 1950 RPM AND CONVERTER OUT TEMPERATURE 220-225°F.



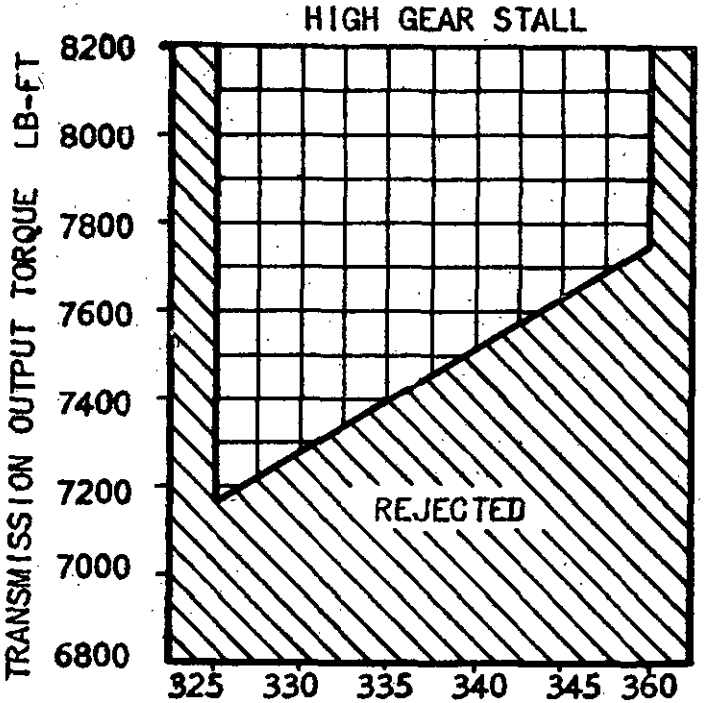
TRANSMISSION OUTPUT SPEED(RPM) AT 2200 RPM AND CONVERTER OUT TEMPERATURE 220 - 225° F.

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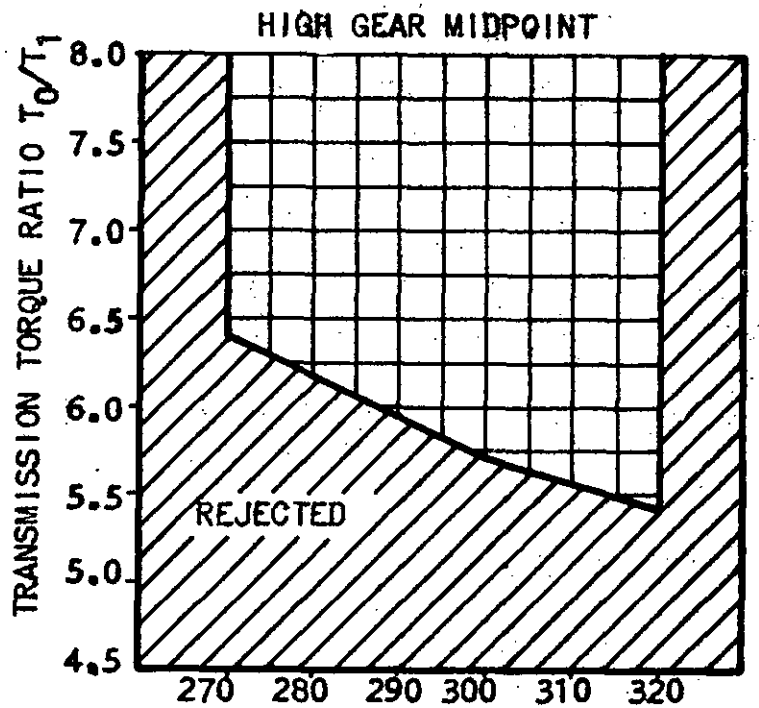
ACCEPTANCE LIMITS
FIGURE 2

FREE RUN

GEAR RANGE	MAXIMUM INPUT TORQUE	ACTUAL INPUT TORQUE
HIGH	37	
LOW	35	
REV.	31	
NEU.	37	



TRANSMISSION INPUT TORQUE--FT-LB AT 2700 RPM INPUT SPEED AND CONVERTER OUT TEMPERATURE 200- 225°F.



TRANSMISSION OUTPUT SPEED--RPM, AT 2500 RPM INPUT SPEED AND CONVERTER OUT TEMPERATURE 200 - 225°F.

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